Note

For example, if we consider n currents in one cycle of AC, namely i_1, i_2, ... i_n, then RMS value is given by

$$I_{RMS} = \sqrt{\frac{\text{Sum of squares of all currents over one cycle}}{\text{Number of currents}}}$$

$$I_{RMS} = \sqrt{\frac{i_1^2 + i_2^2 + ... + i_n^2}{n}}$$

DO YOU KNOW?

For common household appliances, the voltage rating and current rating are generally specified in terms of their RMS value. The domestic AC supply is 230V, 50 Hz. It is the RMS or effective value. Its peak value will be $V_m = \sqrt{2}\ V_{rms} = \sqrt{2} \times 230 = 325 V$.

EXAMPLE 4.18

Write down the equation for a sinusoidal voltage of 50 Hz and its peak value is 20 V. Draw the corresponding voltage versus time graph.

Solution

$f = 50 Hz$; $V_m = 20\ V$

Instantaneous voltage, $\upsilon = V_m \sin \omega t$

$= V_m \sin 2\pi \nu t$

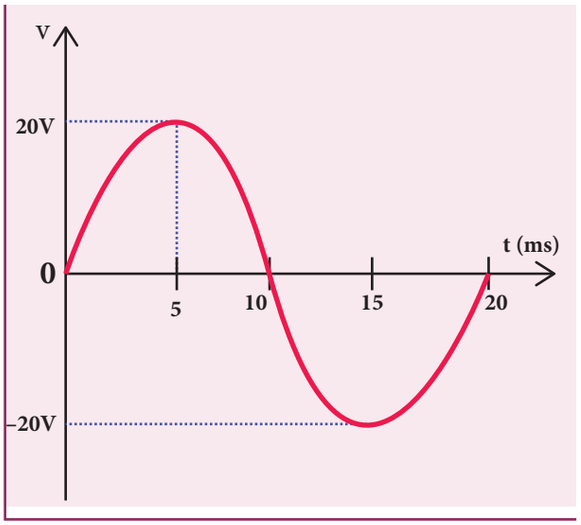
$= 20 \sin(2\pi \times 50)\,t = 20 \sin(100 \times 3.14)\,t$

$\upsilon = 20 \sin 314\,t$

Time for one cycle, $T = \frac{1}{f} = \frac{1}{50} = 0.02\,s$

$= 20 \times 10^{-3} s \quad = 20\,ms$

The wave form is given below.



EXAMPLE 4.19

The equation for an alternating current is given by $i = 77\ sin\ 314t$. Find the peak value, frequency, time period and instantaneous value at $t = 2$ ms.

Solution

$i = 77 \sin 314t$; $t = 2 \text{ ms} = 2 \times 10^{-3} \text{ s}$

The general equation of an alternating current is $i = I_m \sin \omega t$. On comparsion,

(i) Peak value, $I_m = 77\,A$

(ii) Frequency, $f = \frac{\omega}{2\pi} = \frac{314}{2 \times 3.14} = 50\ Hz$

(iii) Time period, $T = \frac{1}{f} = \frac{1}{50} = 0.02\ s$

(iv) At $t = 2$ m s,
Instantaneous value,
$i = 77 \sin(314 \times 2 \times 10^{-3})$
$i = 45.24\,A$

Phasor and phasor diagram

Phasor

A sinusoidal alternating voltage (or current) can be represented by a vector which rotates about the origin in anti-clockwise

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